



DEEP DIVERS

Alvin is one of several similar submersibles that include the Japanese *Shinkai 6500*, the two Russian *Mir* craft, and the Australian *Deepsea Challenger*, designed to make a manned descent to the deepest part of the Pacific Ocean.

◀ DEEPSEA CHALLENGER

In 2012, film director James Cameron used this submersible to dive 35,787 ft (10,908 m) to the bottom of the Mariana Trench.

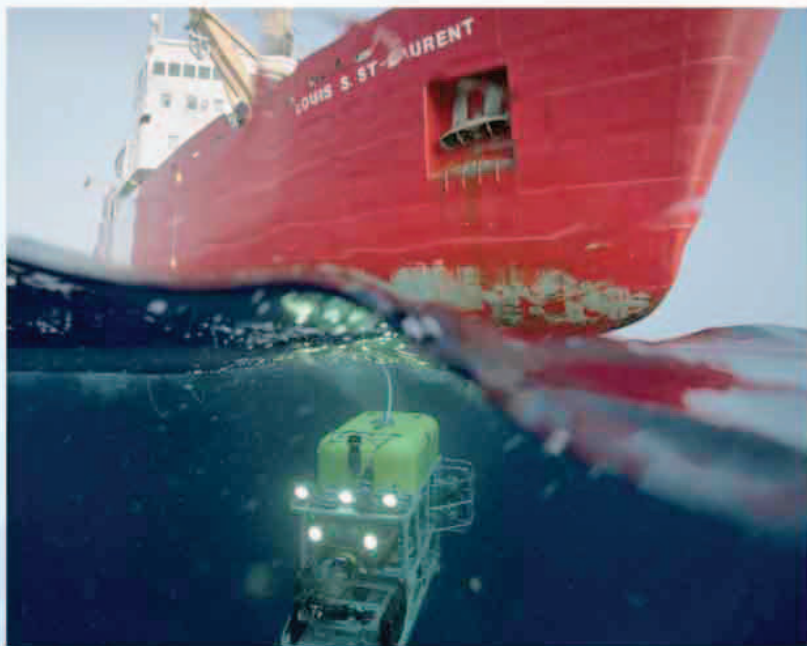
The tall upper part of the Deepsea Challenger contains its batteries and hi-tech lighting arrays

Two mobile booms controlled by the pilot carry a powerful spotlight and a 3-D camera

The pressure-proof cabin is a steel sphere 39 in (100 cm) wide—just big enough for one cramped pilot

▶ HIDDEN WORLD

*This view through the porthole of *Alvin* shows its mechanical arm in action. It is sampling the minerals pouring out of a black smoker on the Juan de Fuca Ridge on the floor of the northeastern Pacific.*



REMOTE VIEW

Only very few manned submersibles have been built because making them safe for the crew is very expensive. It is often easier to use a remotely operated vehicle (ROV) controlled via a video link to a screen on a mother ship. These craft can also explore shipwrecks and other places that are too dangerous for manned vehicles to visit.

NEW DISCOVERIES

Without deep-sea submersibles, we would have little idea of what the deep ocean floor is like and what lives there. In the 1970s, for example, scientists aboard *Alvin* were the first to discover the black smokers that erupt on mid-ocean ridges, and the first to see and collect the amazing wildlife that lives around them.

